Roadmap and Intro to Machine Learning

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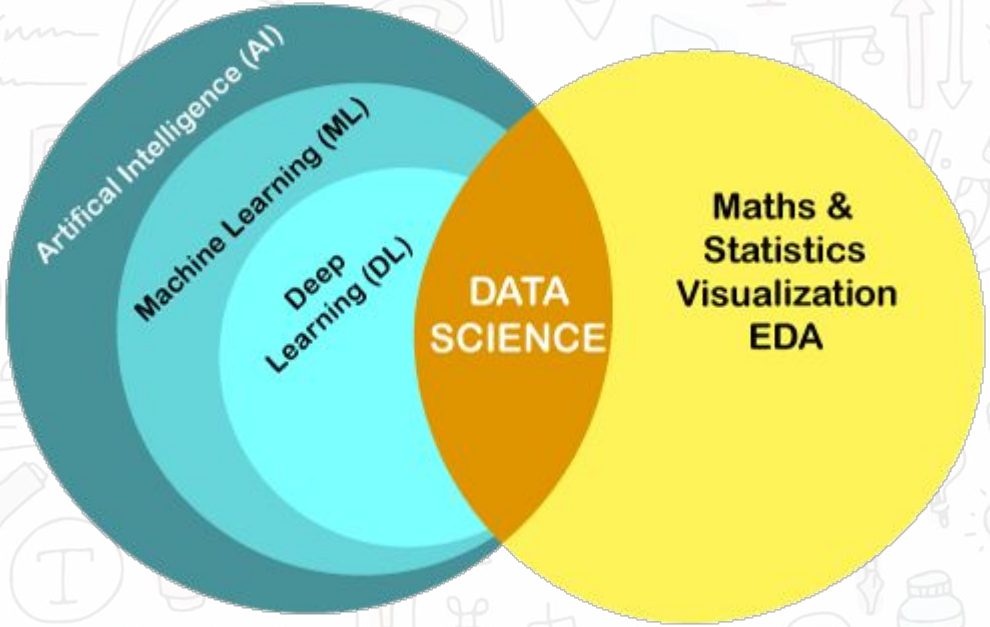
Data Science Vs AI/ML

➡️ Data science is about turning **raw data into useful knowledge.**

➡️Artificial intelligence teaches computers to act and decide like humans



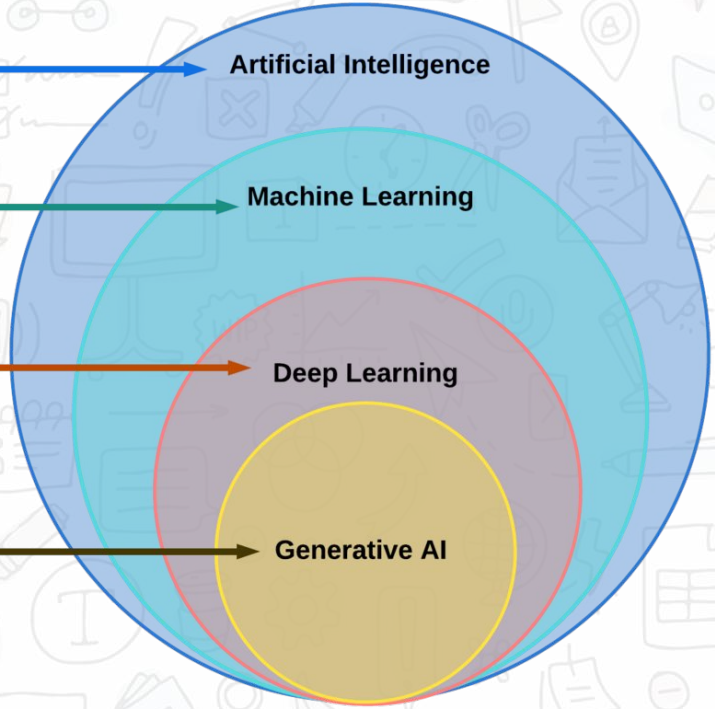
The AI landscape

Artificial Intelligence
AI involves techniques that equip computers to emulate human behavior, enabling them to learn, make decisions, recognize patterns, and solve complex problems in a manner akin to human intelligence.

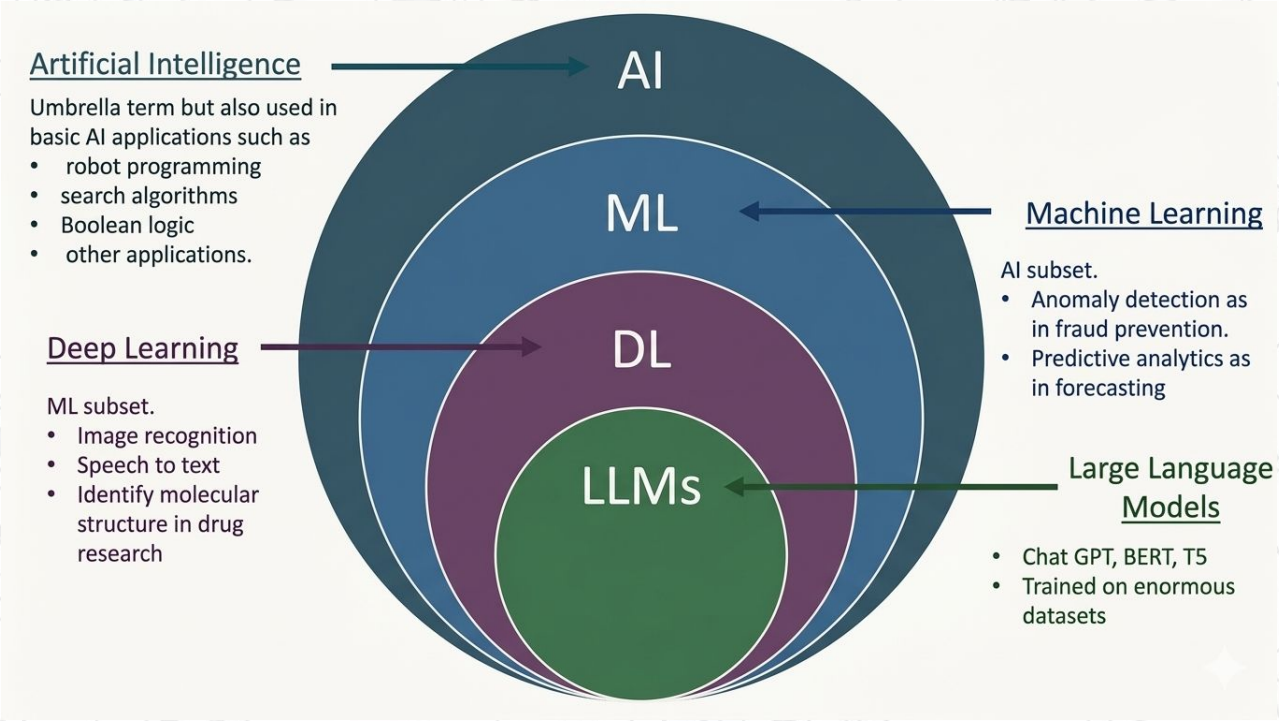
Machine Learning
ML is a subset of AI, uses advanced algorithms to detect patterns in large data sets, allowing machines to learn and adapt. ML algorithms use supervised or unsupervised learning methods.

Deep Learning
DL is a subset of ML which uses neural networks for in-depth data processing and analytical tasks. DL leverages multiple layers of artificial neural networks to extract high-level features from raw input data, simulating the way human brains perceive and understand the world.

Generative AI
Generative AI is a subset of DL models that generates content like text, images, or code based on provided input. Trained on vast data sets, these models detect patterns and create outputs without explicit instruction, using a mix of supervised and unsupervised learning.



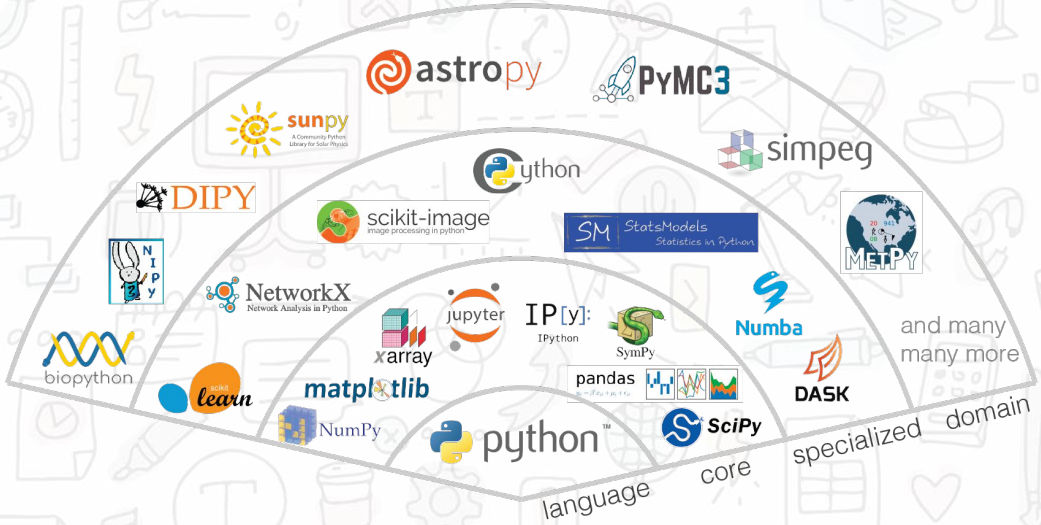
The AI landscape



1. Programming Foundations: Learn Python

Why Python?

- Most AI tools use Python
- Beginner friendly
- Huge ecosystem

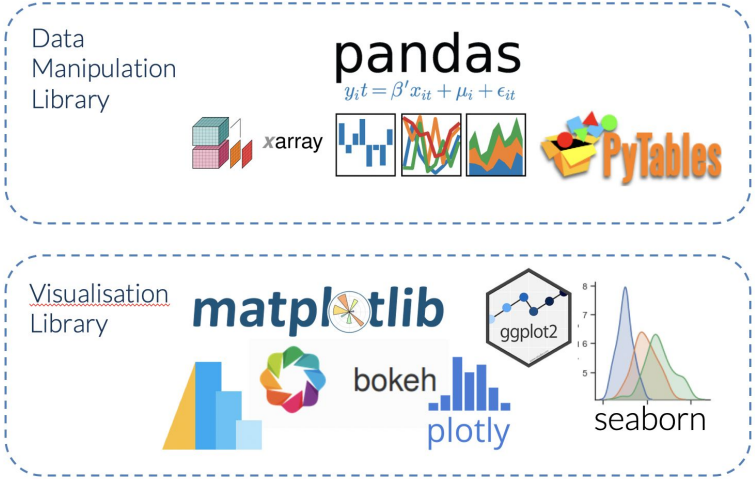


2. Learn Data Handling

AI requires working with data.

Learn:

- NumPy
- Pandas
- Data cleaning
- Dat Visualization

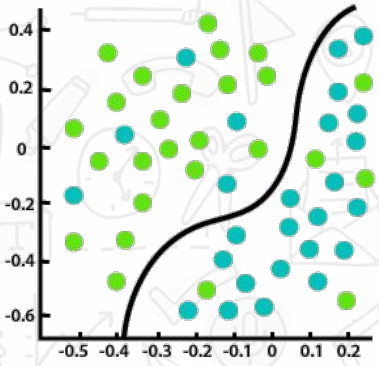


3. Understand Machine Learning Basics

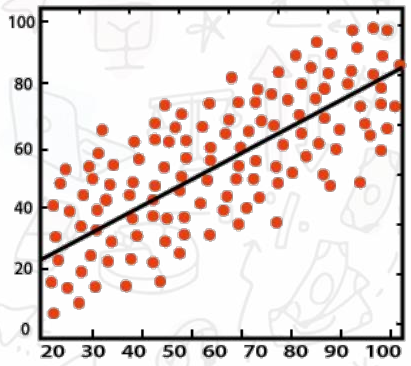
Machine learning is a subset of **artificial intelligence** that enables computer systems to **learn from data.**

Understand concepts:

- Supervised and Unsupervised learning
- Classification and Regression
- Overfitting / Underfitting
- Train/Test split
- Evaluation metrics (Accurcary, F1 etc)



Classification



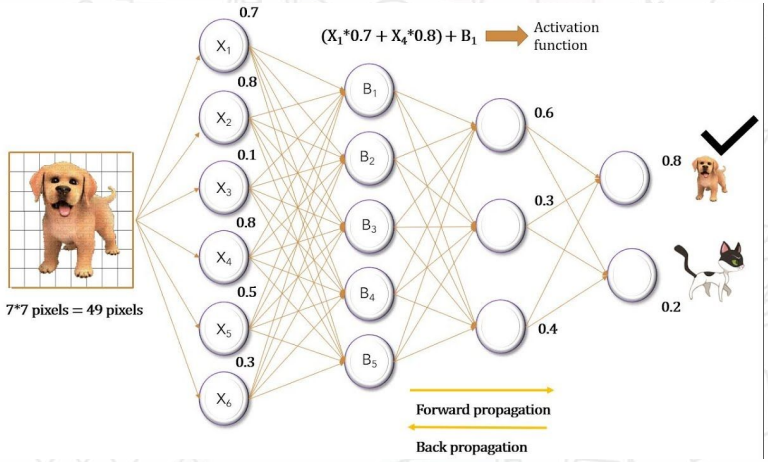
Regression

4. Understanding Deep Learning Basics

Deep Learning is a **subset of ML** that uses **neural networks with many layers** to learn patterns from large amounts of data.

Understand concepts:

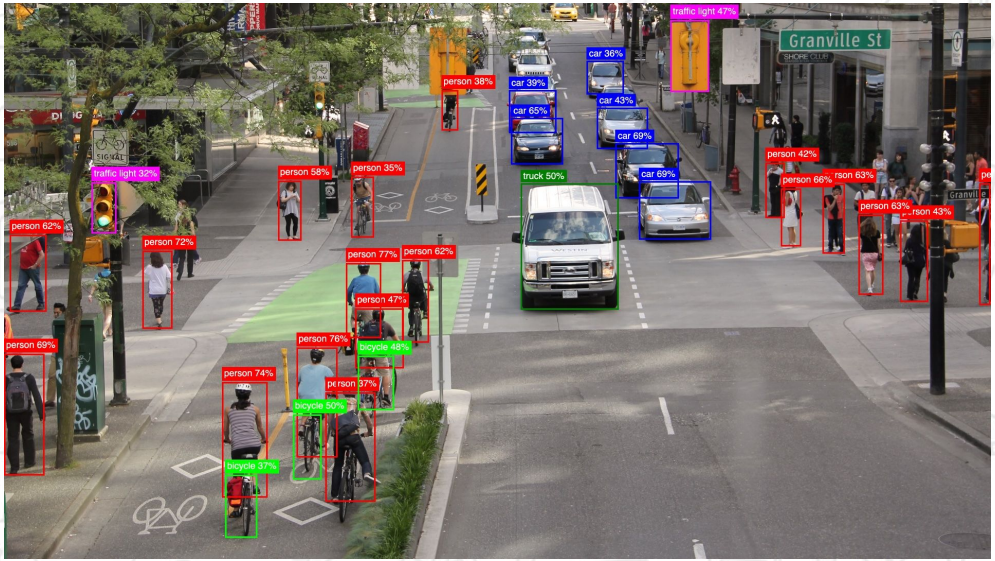
- Neural Networks
- Layers and Activations
- Gradient Descent (How model learns)
- Backward and forward propagations



5. Understanding Computer Vision

Key concepts:

- Image Processing
- Convolutional Neural Network
- Image Classification
- Imag Segmentation
- Object Detection

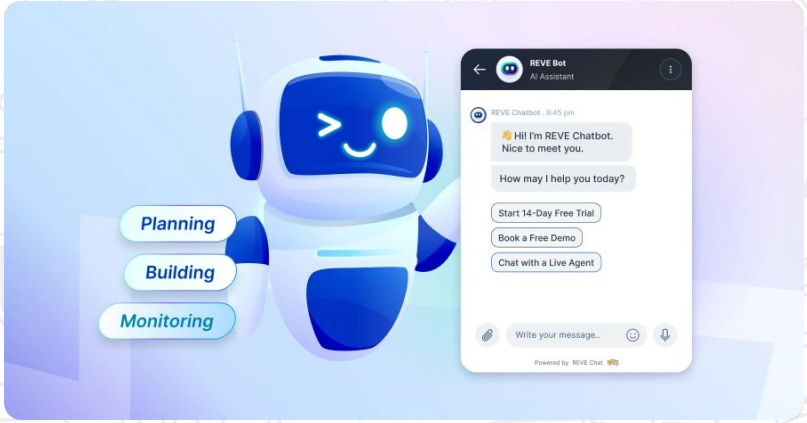


Computer Vision

6. Natural Language Processing Basics

Key Concepts:

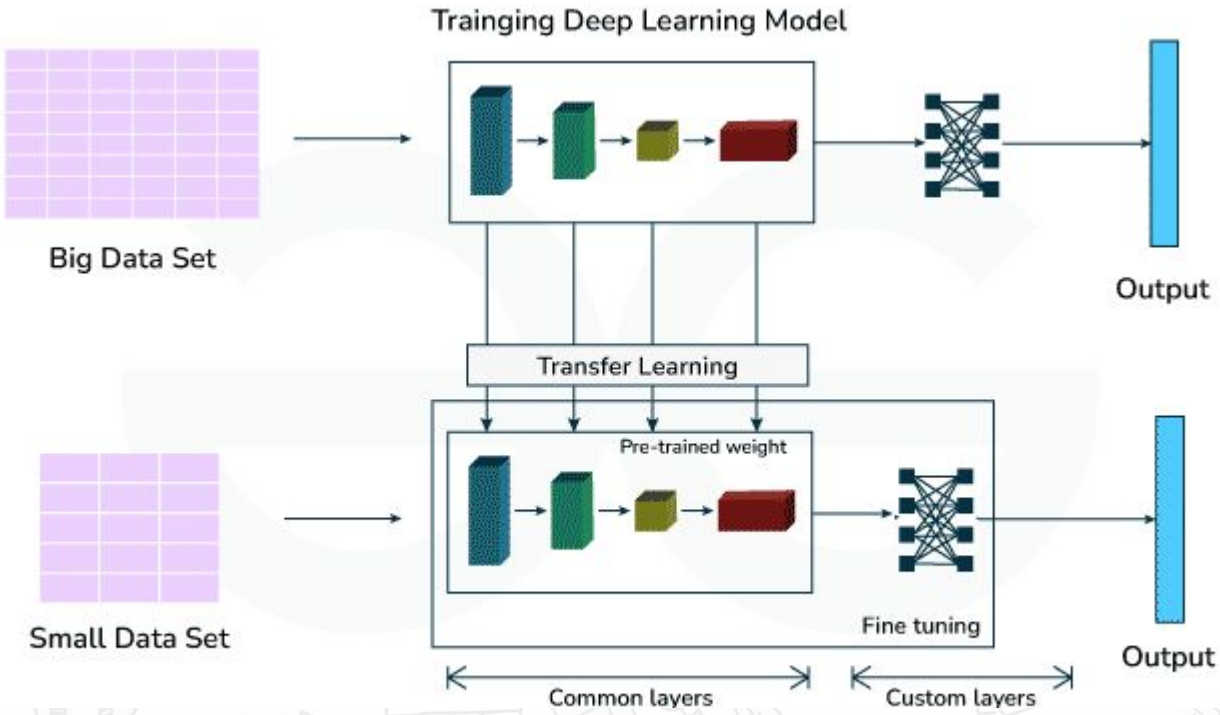
- Text Processing
- Language Models
- RNN and Transformers
- Text Embedding
- Sentiment Analysis
- Text Generation



spaCy

Hugging Face

Model Fine Tuning



7. Learn Modern AI (LLMs)

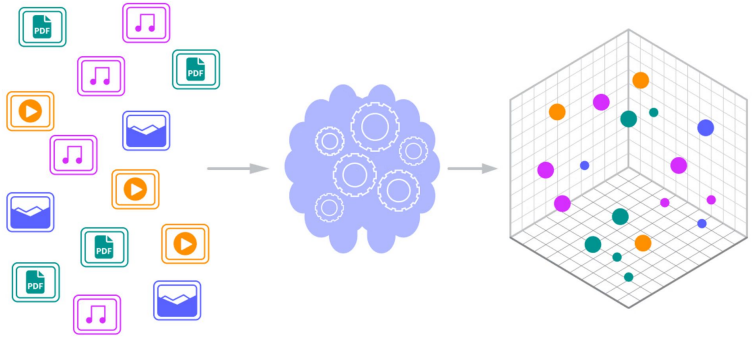
Today AI engineering heavily involves:

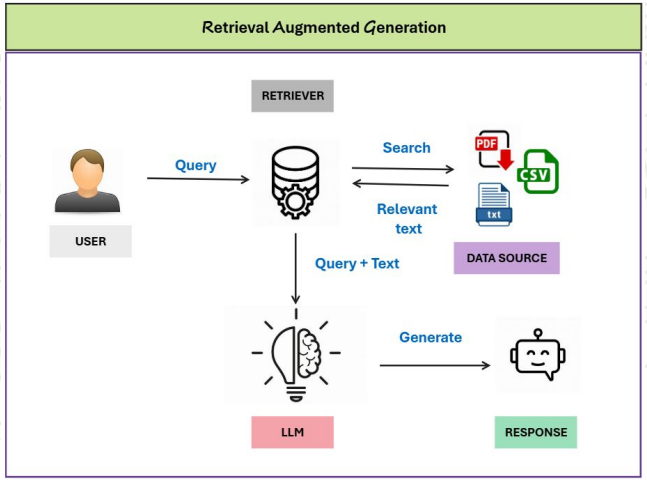
Large Language Models

Examples: GPT, Gemini, Claude, Llama, Qwen

Learn:

- Prompt engineering (React, Structured O/P)
- API usage
- Embeddings
- Vector databases
- RAG (Retrieval Augmented Generation)





How LLM Works

STEP 1 — PREDICTING WORD 1

The cat sat on the **mat**

TOP CANDIDATES CONSIDERED

mat	41%	CHOSEN
floor	24%	
roof	17%	
stairs	11%	
windowsill	7%	

✓ Chosen: "mat" — "mat" wins — it learned this phrase from millions of books and websites. "The cat sat on the mat" is very common in English.

→ Predict next word

↺ Start over

How LLM Works

Probabilities, not certainty

The model scores every word in its vocabulary (50,000+) and picks the most likely one based on context.

Learned from text

These probabilities come from training on billions of books, articles, and websites — it learned what words follow other words.

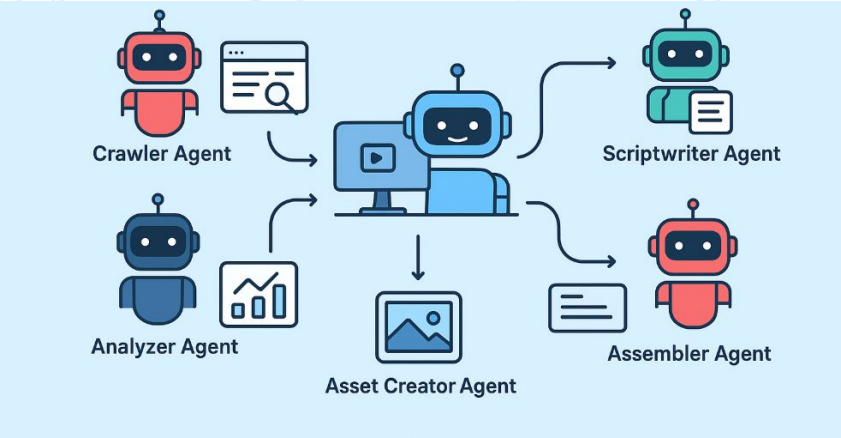
Autoregression

Each new word is fed back as input. A 200-word reply means running this prediction loop 200 times.

8. Working with Agents

Key Concepts:

- Autonomy
- Function Calling
- Tool Usage
- Reasoning & Planning
- Model Context Protocol



crewai

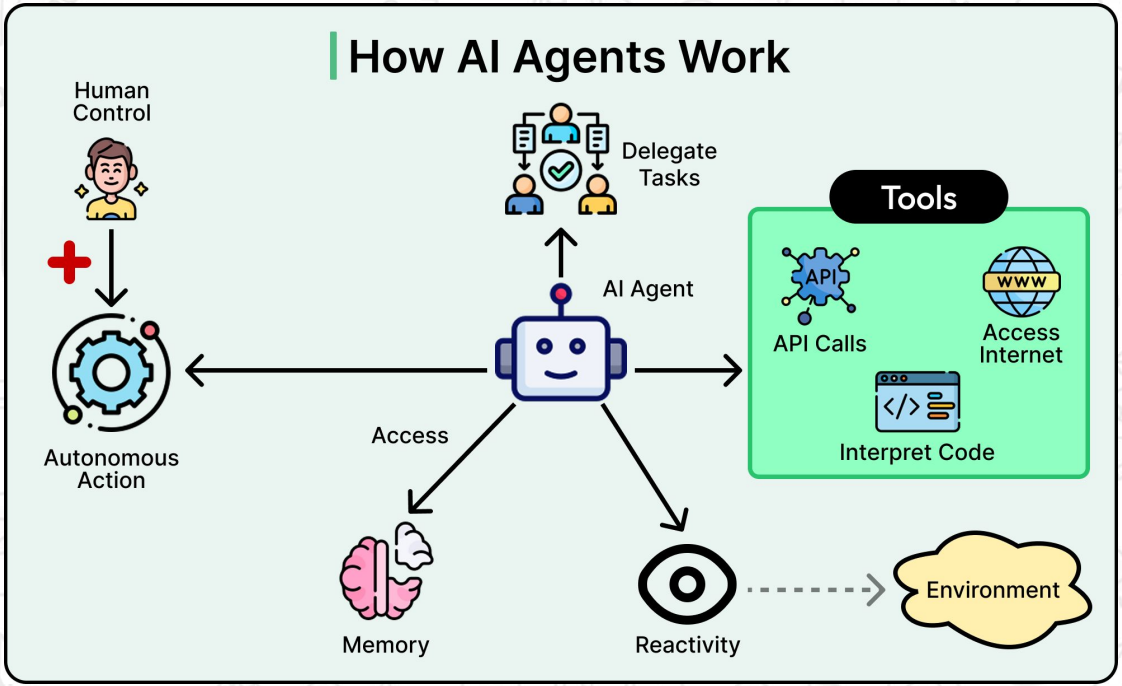
LlamaIndex

LangGraph

Working with Agents

If **LLM = brain** - Thinks, reasons, generates text, But **cannot act on its own**

Then, **Agent = brain + body + tools + decision-making loop**

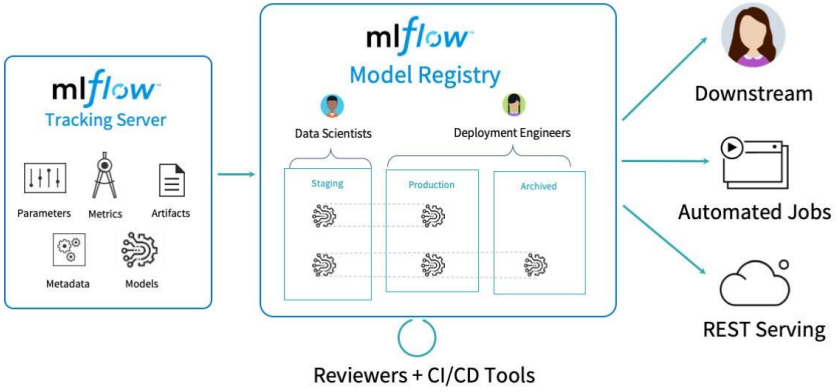


8. MLOps & Production Engineering

Employers hire engineers who can **ship reliable AI Systems**, not just notebooks..

Understand concepts:

- Fastapi
- Git and Github/Gitlab
- Docker
- Model serving, and monitoring



How to Succeed

- **Be present in the live class**. Actively participating will help you stay on track.
- **Don’t rely on shortcuts.** Focus on learning the fundamentals properly.
- **Ask questions.** Never hesitate to seek help when you need it.
- **Stay consistent**. Keep up with the course lessons and follow along.
- **Practice regularly**. Practice truly makes perfect.

Intro to Machine Learning

What is Machine Learning?

Machine learning is a subset of artificial intelligence (AI) that enables systems to **learn from data**, **identify patterns**, and **make decisions or predictions** without being explicitly programmed for every specific rule

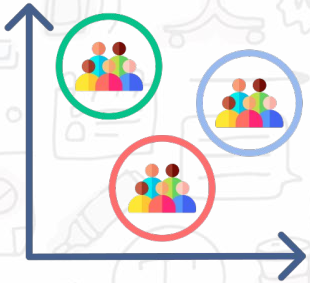
Think of it like this: instead of giving the computer exact instructions, you **show it examples**, and it figures out the patterns on its own.

Type of Machine Learning

Machine Learning

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

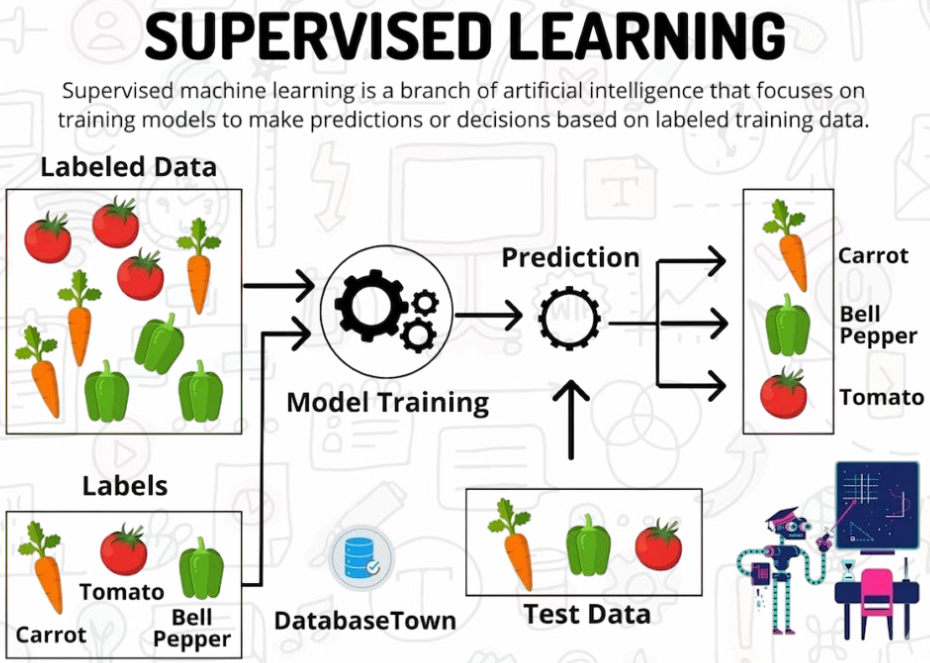
Labels required



No labels required

Learn from mistakes

Supervised Learning

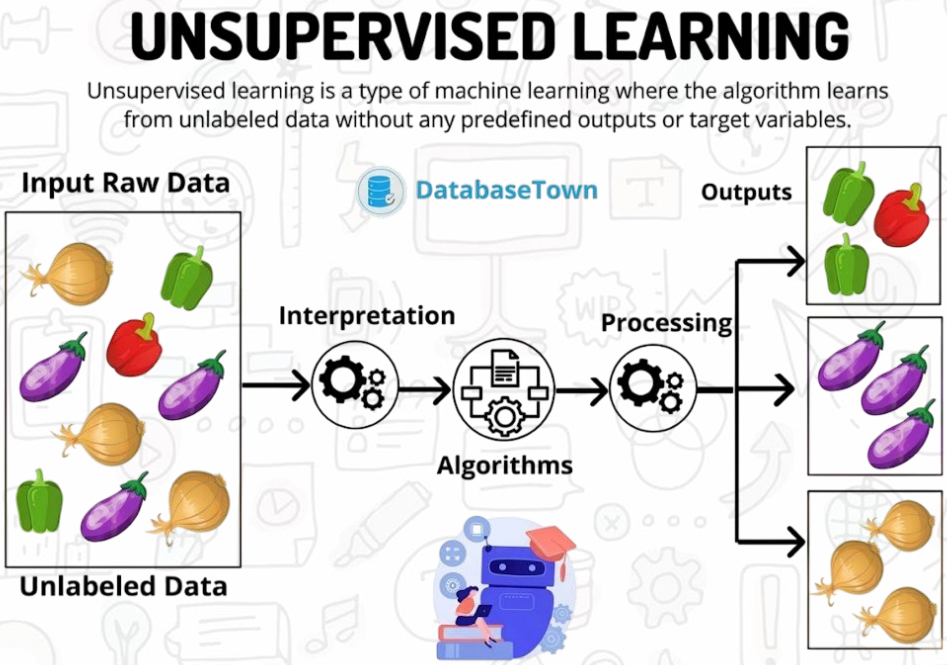


Supervised Learning

Supervised Learning Applications:

- **Email spam detection** – Classifying emails as spam or not spam.
- **Predicting house prices** – Using features like size, location, and rooms to estimate price.
- **Medical diagnosis** – Predicting diseases from patient symptoms or test results.
- **Credit scoring** – Deciding whether a person should get a loan based on financial history.

Unsupervised Learning

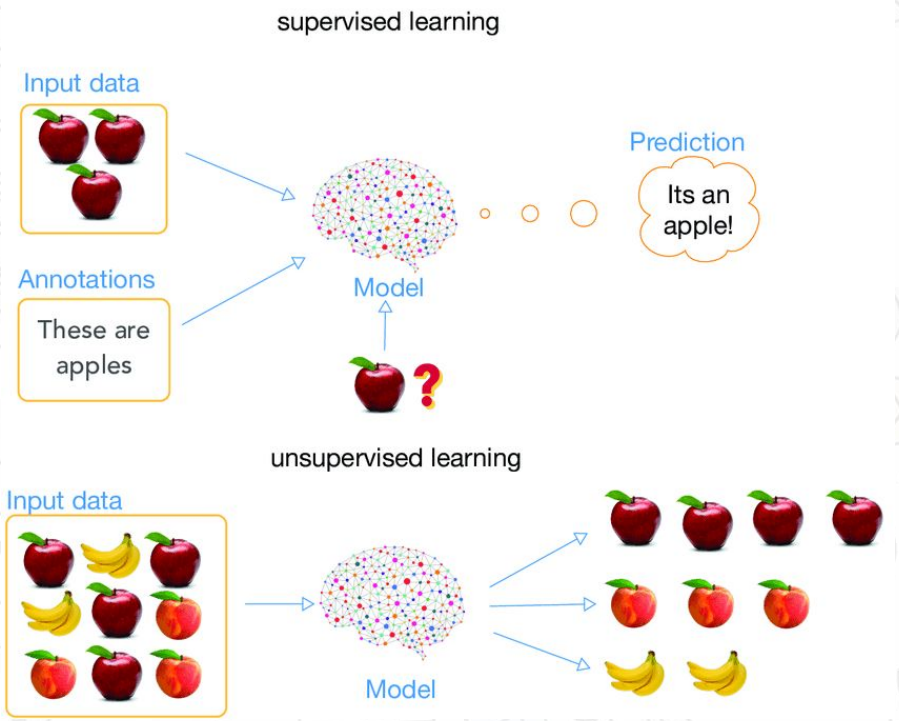


Unsupervised Learning

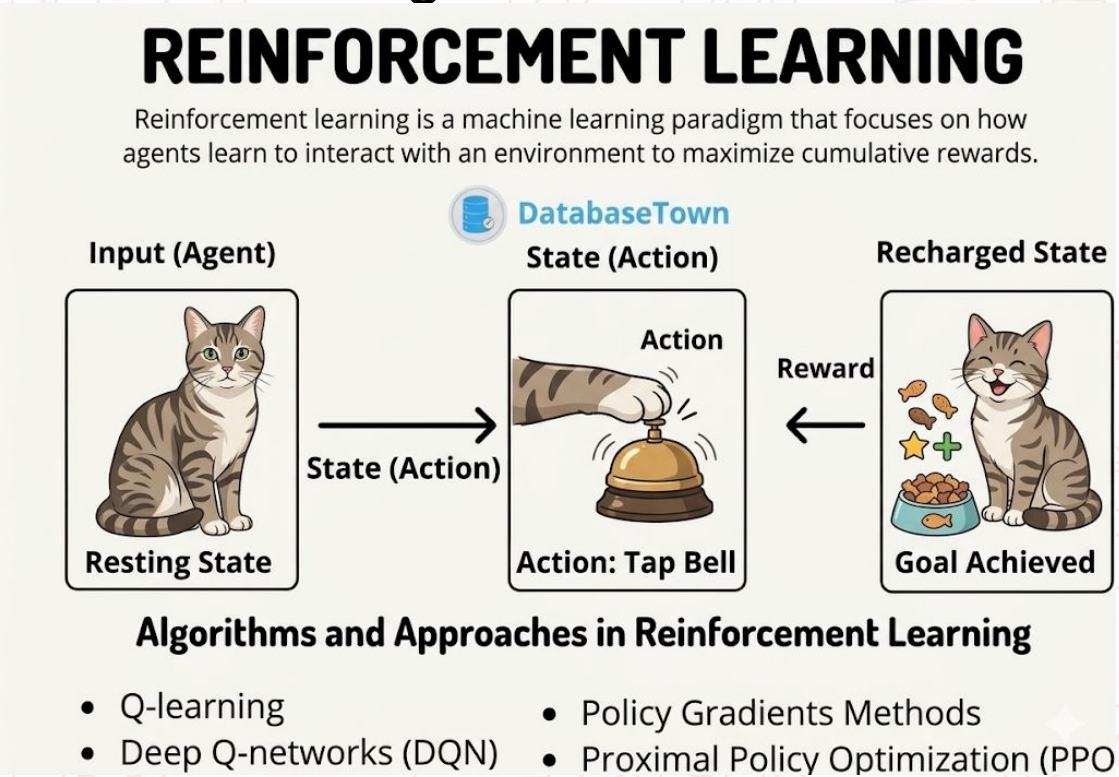
Unsupervised Learning Applications:

- **Customer segmentation** – Grouping customers with similar buying habits for marketing.
- **Market basket analysis** – Finding products that are often bought together (association rules).
- **Anomaly detection** – Detecting unusual patterns, like fraud in transactions.
- **Document clustering** – Organizing news articles or documents into topic-based groups.

Supervised vs Unsupervised



Reinforcement Learning



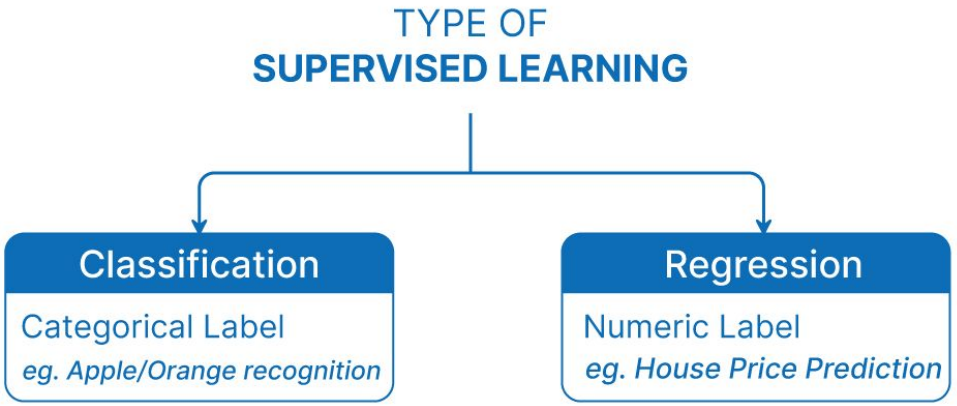
Reinforcement Learning

Reinforcement Learning Applications:

- **Game playing** – Teaching AI to play games like **Chess, Go, or video games** (e.g., AlphaGo, OpenAI Five for Dota 2).
- **Robotics** – Teaching robots to **walk, pick objects, or navigate** in complex environments.
- **Recommendation systems** – Optimizing recommendations dynamically based on **user interactions**.
- **Autonomous vehicles** – Training self-driving cars to **make decisions in traffic** safely.
- **Resource management** – Optimizing **energy usage or traffic signals** for efficiency.

Teaching a Robot how to Walk

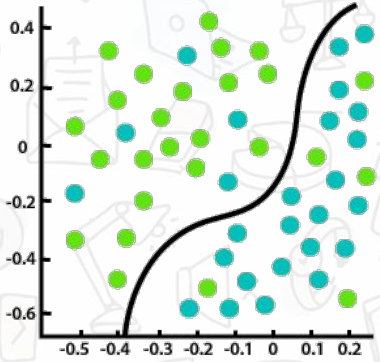
Type of Supervised Learning



Classification

Classification

- **Goal:** Predict a **category or class**.
- **Applications:**
  - Email spam detection (spam or not spam).
  - Diagnosing diseases (disease A, disease B, or healthy).
  - Sentiment analysis (positive, negative, neutral).
  - Handwriting or image recognition (digits, letters, objects).

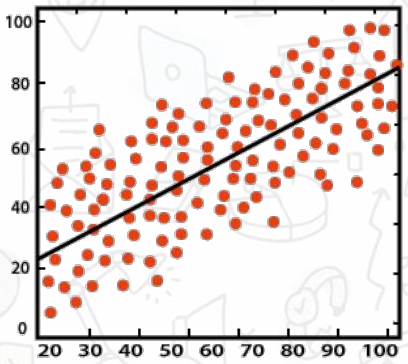


Classification

Regression

Regression

- **Goal:** Predict a **continuous number**.
- **Applications:**
  - Predicting house prices based on size, location, and rooms.
  - Forecasting stock prices or sales.
  - Estimating temperature or weather conditions.



Regression

Classification Vs Regression

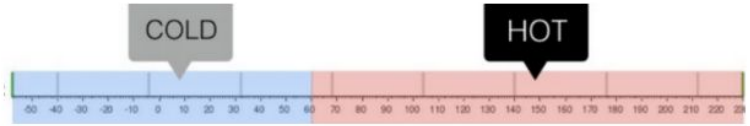
Regression

What will be the temperature tomorrow?

Fahrenheit

Classification

Will it be hot or cold tomorrow?



Fahrenheit

Features and Labels

Features (Input variables)

- **Features** are the **input data** used to make predictions
- They describe the **properties or characteristics** of each example
- Usually denoted as **X**

📌 **Example:**

If you're predicting house prices:

- Size (sq ft)
- Number of bedrooms
- Location
  👉 These are **features**

Labels (Target / Output)

- **Labels** are the **correct answers** you want the model to learn
- They represent what you're trying to predict
- Usually denoted as **y**

📌 **Example:**

- House price
  👉 This is the **label**
- **Labels** are also called **"targets"**

Features and Labels

Regression Dataset

Features

Label

Rows

Size	Beds	Baths	Zip	Price
1100	1	1	64576	1.29
1900	3	1.5	78321	2.14
2800	3	3	98712	3.10
3400	4	3.5	25721	3.75

Columns

Features and Labels

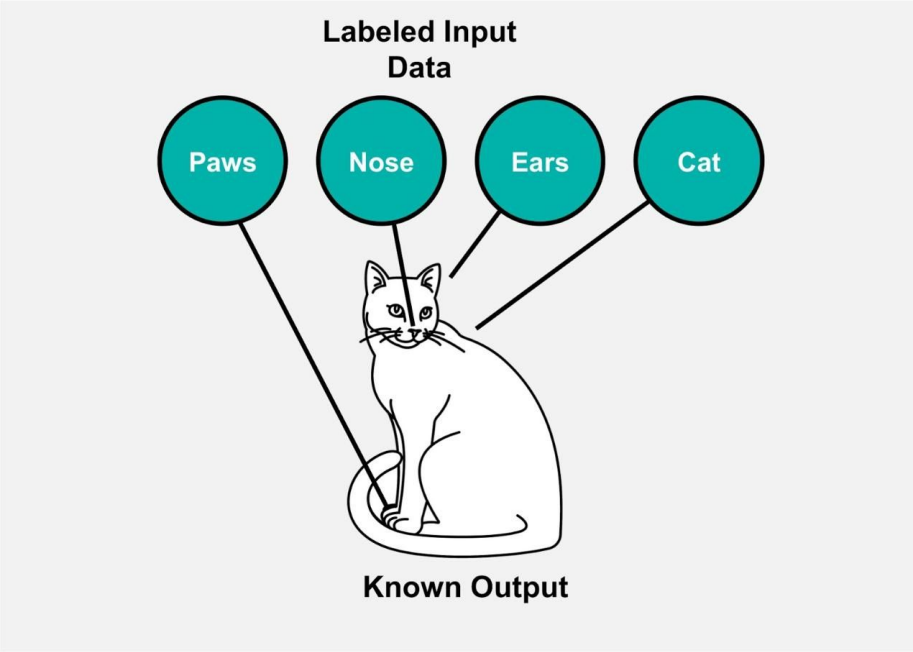
Classification Dataset

Features | label

Sleep Quality	Stress	Weight	Migraine
Poor	Yes	97	Yes
Good	No	78	No
Poor	No	47	Yes
⋮	⋮	⋮	⋮
Moderate	No	76	Yes

Features and Labels

In this image, what are the features and what is the label?



Thank you!